EXP.NO:07

G-COOKIES PROBLEM

PROBLEM STATEMENT:

Assume you are an awesome parent and want to give your children some cookies. But, you should give each child at most one cookie.

Each child i has a greed factor g[i], which is the minimum size of a cookie that the child will be content with; and each cookie j has a size s[j]. If s[j] >= g[i], we can assign the cookie j to the child i, and the child i will be content. Your goal is to maximize the number of your content children and output the maximum number.

Example 1:

Input:

3

1 2 3

2

1 1

Output:

1

Explanation: You have 3 children and 2 cookies. The greed factors of 3 children are 1, 2, 3.

And even though you have 2 cookies, since their size is both 1, you could only make the child whose greed factor is 1 content.

You need to output 1.

Constraints:

1 <= g.length <= 3 * 10^4

0 <= s.length <= 3 * 10^4

1 <= g[i], s[j] <= 2^31 - 1

CODE:

```c
#include<stdio.h>
#include<stdlib.h>

int cmp(const void *a, const void *b){
    return *(int *)a - *(int *)b;
}

int main(){
    int n, m, i = 0, j = 0, count = 0;
    scanf("%d", &n);
    int g[n];
    for(int i=0;i<n;i++)
        scanf("%d", &g[i]);
    scanf("%d", &m);
    int s[m];
    for(i=0;i<m;i++)
        scanf("%d", &s[i]);
    qsort(g, n, sizeof(int), cmp);
    qsort(s, m, sizeof(int), cmp);

    i=j=0;

    while(i<n && j<m){
        if(s[j] >= g[i]){
            count++;
            i++;
        }
        j++;
    }
    printf("%d", count);
    return 0;
}
```

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OUTPUT:



	Input	Expected	Got	
✔	2 1 2 3 1 2 3	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

RESULT:

The code is executed successfully.